Zhuohan Wang

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Education

Harvard University — Cambridge, MA
M.S. in Computational Science and Engineering — Sep 2025 – Present

City University of Hong Kong — Hong Kong
BEng in Computer and Data Engineering — Aug 2021 – Jun 2025
- **GPA: 3.95** (Rank **1st** / 69); **Summa Cum Laude**; HKSAR Government Scholarship

Professional Experience

Mitsubishi Electric Research Laboratories (MERL) — Cambridge, MA
Research Intern; Mentors: Dr. Pu (Perry) Wang, Dr. Anoop Cherian — May 2026 – Sep 2026
Publications: **3 papers under review at ICLR 2027**, 2 as first author
- Tested whether **vision-language models** can use the image to correct false visual claims in their own reasoning, designing a controlled replay evaluation which shows that they largely fail to, even with the evidence still in context.
- Found that copying RGB chain-of-thought into **radar and audio** students hurts, and designed a method that re-derives each rationale against the target sensor first, beating naive distillation on every backbone tested.
- Defined what a benchmark for evidence-grounded sensor reasoning about human activity should measure, separating **understanding** from **reasoning**; designed the question types and built the question-generation pipeline, then handed it off to the data-collection and evaluation teams.

SenseTime Research — Hong Kong
Algorithm Researcher Intern; Advisor: Dr. Litong Feng — Jul 2024 – Sep 2025
- **Improving Instruction-Following in Proprietary LLM:** Participated in data workflows and optimization for **SenseChat**, contributing to **SFT**, **DPO**, and **RL** stages. Took independent responsibility for improving **instruction-following ability**, achieving a **top-five ranking** on the **SuperCLUE** benchmark (bi-monthly updates), and driving additional gains on **LiveBench** and **IFEval** compared to leading models such as Claude and OpenAI's latest releases.
- **Dataset and benchmark construction:** Contributed to the design and construction of datasets and benchmarks for tasks such as **safety**, Sudoku, and role-playing, achieving consistent gains of **2–3 percentage points** across multiple evaluation tracks.
- Engaged in **multi-agent research** aimed at enhancing **hard mathematical reasoning** capabilities of LLMs.

SHANGHAI-FANUC Robotics — Shanghai, China
Machine Vision Engineer Intern — Jul 2022 – Aug 2022
- **Calibration and segmentation accuracy:** Improved calibration and segmentation accuracy by developing and refining computer-vision algorithms.
- **Visualization tools for decision support:** Integrated tailored visualization tools for calibration and segmentation, aiding decision-making and stakeholder communication.

Research Experience

LLM Reasoning Evaluation and Efficient Decoding — Cambridge, MA
Independent Researcher, Harvard University — Sep 2025 – Present
Publications: **2 accepted at NeurIPS 2026** (1 Spotlight, 1 recommended for oral) [1][3]; **2 preprints** [4][5] and 1 more paper under review
- Led **OracleLadder** to locate where LLM math reasoning breaks, giving models increasing **oracle help** and grading answers with a symbolic verifier; in six models (8B–671B), the **composition gap** (every step solved, whole problem failed) was the largest failure (**33–48%** of problems).
- Tested how well **process reward models**, validated as step checkers, choose branches in test-time search by scoring each choice on final correctness, showing that step correctness is a weak signal for branch selection.
- Helped formulate the idea of **D-PACE**, a **speculative decoding** drafter loss that weights each position by its contribution to accepted length; then, as co-first author, extended it to weight positions by the true acceptance probability.

Causal Sufficiency and Necessity for Chain-of-Thought Reasoning — London, UK
Research Assistant; Advisor: Dr. Mengyue Yang — Jan 2025 – Jul 2025
Publications: **1 accepted at NeurIPS 2025**, co-first author [2]
- Introduced a **causal framework** for **Chain-of-Thought** reasoning (**sufficiency & necessity**) to rigorously identify **essential reasoning steps**.
- Developed a **PNS-guided pruning** method that **improves accuracy** on easier benchmarks while **significantly reducing**

token usage, and on harder benchmarks achieves **20–60% fewer tokens/steps** while maintaining **comparable accuracy**.

Parameter-Efficient Fine-Tuning for Large Language Models Hong Kong
Research Assistant; Advisor: Prof. Lai-Man Po Jun 2023 – Mar 2024
Publications: **1 accepted at ICONIP 2024** [6]

- Conducted research on **parameter-efficient fine-tuning (PEFT)** for **LLMs**, developing **LoRA** and **GaLore** variants that **improved convergence**, **reduced memory footprint and training cost**, and **enhanced benchmark performance** across multiple tasks, including **SBoRA**.
- Proposed and developed **Colore**, a novel PEFT method for **LLMs** achieving **2× training speed** over GaLore under specific settings, while maintaining **competitive benchmark performance**.

Large-Scale Entity Matching and Embedding-based Record Linkage Hong Kong
Research Engineer; Advisor: Prof. Rengong Zhang Jun 2023 – Jul 2024

- Contributed to a **government-funded research project** on large-scale **entity matching of company names**. Built a system operating on **terabyte-scale data**, integrating heterogeneous databases and leveraging **embedding models** to deliver an **automated, high-accuracy pipeline**.
- Developed a **data pipeline** for an **HR consulting project** that mapped project names to ranking categories. Applied **clustering techniques** and **LLMs** to improve **matching precision** and deliver a **scalable automated solution**.

Selected Publications (* equal contribution)

Reasoning

[1] Solving Every Step Is Not Enough: Milestone Oracles Reveal a Composition Gap in LLM Math Reasoning
Z. Wang (project lead), H. Ma, T. Wu, Y. Duan, Z. Liao, J. Yu
NeurIPS 2026, Evaluations and Datasets Track (Recommended for oral). [arXiv] [Paper] [Code]

[2] Causal Sufficiency and Necessity Improves Chain-of-Thought Reasoning
X. Yu*, **Z. Wang***, L. Yang, H. Li, A. Liu, X. Xue, J. Wang, M. Yang
NeurIPS 2025. [arXiv] [Code]

Efficient Training and Inference

[3] D-PACE: Dynamic Position-Aware Cross-Entropy for Parallel Speculative Drafting
T. Wu, Y. Yao, Z. Qi, H. Zheng, C. Zhang, **Z. Wang**, H. Ma, Z. Liao, H. Lakkaraju, J. Li, Y. Du
NeurIPS 2026 (Spotlight). [Paper] [Code]

[4] Beyond the Point Mass: Acceptance-Derived Weights and Distributional Losses for Speculative Decoding
T. Wu*, **Z. Wang***, J. Geuter, Y. Yao, W. Jiang, C. Zhang, J. Li, M. Yu
Preprint, 2026. [arXiv] [Code]

[5] LoopKV: Loop-Aware KV Cache Compression for Serving Looped Transformers
T. Zhou*, Y. Zhang*, T. Wu*, **Z. Wang***, W. Nixon, J. Yang
arXiv:2611.04137, 2026. [arXiv]

[6] SBoRA: Low-Rank Adaptation with Regional Weight Updates
L.-M. Po, Y. Liu, H. Wu, T. Zhang, W.-Y. Yu, **Z. Wang**, Z. Jiang, K. Li
ICONIP 2024. [arXiv] [Code]

Skills & Languages

Programming: Python (primary), C++, Java, SQL, R, MATLAB, Assembly

ML/AI: PyTorch, scikit-learn; SGLang, vLLM; OpenAI APIs; **Cloud/DevOps:** AWS, Docker, Git

Languages: Native Mandarin & Shanghainese; Proficient English **Interests:** Badminton, fitness, table tennis, jogging, investment, emerging technologies, listening to podcasts